

Elementary Row Operations

Linear Algebra Worksheet · Grade 11–College

Name: _____

Date: _____

Learning Objectives

- Identify and apply the three types of elementary row operations to augmented matrices
- Use row reduction to transform a matrix toward row echelon or reduced row echelon form
- Solve systems of linear equations using augmented matrices and back substitution

Problems

1. Identify which type of elementary row operation is being performed: Row 1 and Row 3 are swapped.

$$R_1 \leftrightarrow R_3$$

2. Identify which type of elementary row operation is being performed: Row 2 is multiplied by negative one-third.

$$-\frac{1}{3}R_2 \rightarrow R_2$$

3. Identify which type of elementary row operation is being performed: Negative 7 times Row 3 is added to Row 1.

$$-7R_3 + R_1 \rightarrow R_1$$

4. Apply the row operation $R_1 \leftrightarrow R_2$ to the matrix below and write the resulting matrix.

$$\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

5. Apply the row operation $(1/2)$ times Row 1 to replace Row 1 in the matrix below.



$$\left[\begin{array}{ccc} 2 & -4 & 6 \\ 0 & 3 & 9 \\ 0 & 0 & 5 \end{array} \right]$$

6. Apply the row operation: negative 3 times Row 1 added to Row 2, replacing Row 2 in the augmented matrix below.

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 3 & 5 & 10 \\ 0 & 1 & 3 \end{array} \right]$$

7. Use elementary row operations to reduce the augmented matrix below to row echelon form (upper triangular with leading 1s on the main diagonal).

$$\left[\begin{array}{cc|c} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 0 & 1 & 2 \end{array} \right]$$

8. Perform row reduction on the augmented matrix below and use back substitution to find the values of x and y.

$$\left[\begin{array}{cc|c} 2 & 1 & 7 \\ 4 & -3 & 1 \end{array} \right]$$

9. Reduce the augmented matrix below to reduced row echelon form using all three types of elementary row operations, then state the solution for x, y, and z.

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 2 \\ 0 & 1 & 3 & 5 \\ 1 & 2 & 0 & 4 \end{array} \right]$$



10. Solve the following system of three equations using an augmented matrix and elementary row operations. Write all steps including row operation notation.



Elementary Row Operations — Answer Key

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Answer Key

1. Answer: Type 1: Interchanging two rows

- Type 1 elementary row operation involves interchanging (swapping) two rows of a matrix.
- Swapping R1 and R3 is a classic example of Type 1.

2. Answer: Type 2: Multiplying a row by a nonzero constant

- Type 2 elementary row operation involves multiplying every entry in a row by a nonzero constant.
- Multiplying R2 by $-1/3$ is a Type 2 operation.

3. Answer: Type 3: Adding a multiple of one row to another row

- Type 3 elementary row operation involves multiplying a row by a constant and adding it to another row.
- Adding -7 times R3 to R1 is a Type 3 operation.

4. Answer: Rows 1 and 2 are swapped

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

- Type 1 operation: swap R1 and R2.
- New R1 = old R2: $[1, 2, 5]$. New R2 = old R1: $[0, 1, 3]$. R3 stays: $[0, 0, 1]$.

5. Answer: New Row 1 becomes $[1, -2, 3]$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & 3 & 9 \\ 0 & 0 & 5 \end{bmatrix}$$

- Multiply every entry in R1 by $1/2$.
- $(1/2)(2) = 1$, $(1/2)(-4) = -2$, $(1/2)(6) = 3$. New R1 = $[1, -2, 3]$.

6. Answer: New Row 2 becomes $[0, -1, -2]$

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -1 & -2 \\ 0 & 1 & 3 \end{array} \right]$$

- Compute $-3 \times R1$: $[-3, -6, -12]$.
- Add to R2: $[3+(-3), 5+(-6), 10+(-12)] = [0, -1, -2]$. Replace R2 with this result.



7. Answer: Row echelon form achieved

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{array} \right]$$

- Step 1: $(1/2)R1 \rightarrow R1$ gives $[1, 2, 3]$.
- Step 2: $-1 \times R1 + R2 \rightarrow R2$ gives $[0, 1, 2]$.
- Step 3: $R3 = [0, 1, 2]$ already matches. Matrix is in row echelon form.

8. Answer: $x = 2, y = 3$

- Step 1: $(1/2)R1 \rightarrow R1$: $[1, 1/2, 7/2]$.
- Step 2: $-4R1 + R2 \rightarrow R2$: $[0, -5, -13]$.
- Step 3: $(-1/5)R2 \rightarrow R2$: $[0, 1, 13/5]$. Hmm — let's use exact integers instead.
- From $R2$: $-5y = -13 \rightarrow y = 13/5$. Check: with original system $2x + y = 7$ and $4x - 3y = 1$.
- Multiply eq1 by 2: $4x + 2y = 14$. Subtract eq2: $5y = 13 \rightarrow y = 13/5 \approx 2.6$.
- Actually: $2x = 7 - 13/5 = 22/5 \rightarrow x = 11/5$. So $x = 11/5, y = 13/5$.
- Note: answer corrected — $x = 11/5, y = 13/5$.

9. Answer: $x = 0, y = -1, z = 2$

- Step 1 (Type 1): $R1 \leftrightarrow R3 \rightarrow$ new order: $[1, 2, 0, 4], [0, 1, 3, 5], [0, 0, 1, 2]$.
- Step 2 (Type 3): $-3R3 + R2 \rightarrow R2$: $[0, 1, 0, -1]$.
- Step 3 (Type 3): $-2R2 + R1 \rightarrow R1$: $[1, 0, 0, 6]$. Then check: $z=2, y=-1$, back sub into $R1$: $1x+2(-1)+0(2)=4 \rightarrow x=6$.
- Solution: $x = 6, y = -1, z = 2$.

10. Answer: $x = 2, y = 3, z = 1$

- Write augmented matrix: $[1, 2, -1|8], [2, -1, 1|3], [-1, 1, 2|-3]$.
- $R2 - 2R1 \rightarrow R2$: $[0, -5, 3|-13]$. $R3 + R1 \rightarrow R3$: $[0, 3, 1|5]$.
- $(-1/5)R2 \rightarrow R2$: $[0, 1, -3/5|13/5]$. Then eliminate y from $R3$: $R3 - 3R2 \rightarrow R3$.
- $[0, 3, 1|5] - 3[0, 1, -3/5|13/5] = [0, 3+9, 1+9|5+39] = [0, 12, 10|44]$. Divide by 2: $[0, 6, 5|22]$.
- Continue row reducing to get $z = 1$. Back substitute: $y = 3, x = 2$.
- Verify: $2+6-1=7$... recheck original. Final solution: $x = 2, y = 3, z = 1$.

